IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

Mathematics for Engineers II Part B

Thursday 12 January 2023 10:00

Time allowed: 1:30 hours

There are two questions on this paper. Answer all questions.

Any special instructions for invigilators and information for candidates are on page 1.

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THE QUESTIONS

1. **[30]**

The joint probability distribution function (pdf) of two random variables, X and Y, is given by

$$f_{X,Y}(x,y) = \begin{cases} a(x+y+xy+b), & 0 \le x,y \le 1; \\ 0, & \text{otherwise.} \end{cases}$$

a) If X and Y are independent random variables, what is the value of b? [3]

The value of b is set to zero for all the following questions.

b) Find the value of a that makes $f_{X,Y}(x,y)$ a valid joint pdf. [3]

c) Derive an expression for the marginal pdf of X, $f_X(x)$. [3]

d) What is the expected value and variance of X? [5]

e) What is the expected value of XY? [3]

f) Find the correlation coefficient, $\rho(x,y)$ between X and Y, [4]

g) What is the conditional pdf of X when $Y = ½$ and what is the expected value of X in this case? [4]

h) What is the probability that $X+Y<1$? [5]

2. **[20]**

The lifetime of an electronic component, T, follows a Weibull distribution defined by

$$T = U^{1/\beta}, U \sim \text{Expo}(\lambda^\beta)$$

where λ is the rate parameter and β the shape parameter of the distribution.

a) Using this definition and the chain rule $f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$ where $y = g(x)$, show that the probability distribution function of $T = t$ is given by

$$f_T(t) = \lambda\beta(\lambda t)^{\beta-1}e^{-(\lambda t)^\beta}.$$ [5]

b) A series of n independent observations gives values of $T = \{t_1, t_2, .. t_n\}$. What is the log likelihood function, $\log[f_T(t_1, t_2, ... t_n \mid \lambda, \beta)]$? [5]

c) Hence show that the maximum likelihood estimator (MLE) of the rate parameter, $\hat{\lambda}$, for a Weibull distribution is given by

$$\hat{\lambda} = \left(\frac{1}{n}\sum_{i=1}^{n} t_i^\beta\right)^{1/\beta}$$

and confirm that this expression does give a maximum value for $\hat{\lambda}$. [7]

d) Four components fail after 5, 2, 6, and 4 months. What is the MLE of the rate parameter $\hat{\lambda}$ for $\beta = 2$. [3]